Inner conjugacy and residual primitivity in the HS supplement

A finite source component using acquired subgroup data

We use the fixed six-page article by Dimitri Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, and its retained 23-page supplement dated 6 October 2003. Diagram numbers below refer to that supplement. This component checks its conjugacy notes, three stronger primitivity annotations, and parenthetic numbers. It does not assert whole-source completion.

The acquired model is $G = \mathrm{HS}$ on 100 points, of order 44352000. The accepted input contains 589 representatives of subgroup classes and 3075 actual maximal-inclusion pairs (K, M), with each parent K one of these representatives. Their completeness and the named ambient model are retained input premises, already acquired by the completed geometry classification. We use the entire maximal-inclusion list, including inclusions outside its separate locally two-transitive sublist. The accepted classification supplies 86 literal families $(P_1, \ldots, P_r)$; the independently accepted source-coordinate bijection supplies the ordered source-to-model type permutation. The accepted parabolic structure packet supplies the group names.

Proposition 1. *All 24 subgroup-pair relations specified by the 18 conjugacy notes listed below hold inside G, with the source's actual type coordinates.*

Proof. For each pair (P, Q) the certificate applies the complete subgroup conjugation algorithm `RepresentativeAction(G,P,Q,OnPoints)`. Every positive result is retained as an actual permutation $x \in G$, and the replay verifies $P^x = Q$. There are eleven positive results. For each of the thirteen negative results the exact complete algorithm returns failure; the replay checks that failure again. Completeness of this exact permutation-group algorithm is an explicit software premise. No claim is inferred merely from equality of group orders or names, nor is conjugation in $\mathrm{Aut}(G)$ substituted for conjugation in G.

The source join fixes both ordered type indices before this test. Thus even repeated abstract names are checked at their actual occurrences. For 4.41 the top pair and bottom pair are each conjugate, and all four cross pairs are nonconjugate. Diagram 5.6 has two separate nonconjugacy claims, for M_{11} and A_7 respectively. □

Name	Conjugate in G	Nonconjugate in G
M_{11}	3.16, 3.30, 4.15	2.4, 4.39, 4.40, 5.6, 5.7
$\mathrm{U}_3(5) : 2$	4.9, 4.10, 4.25, 4.26	3.1
$\mathrm{U}_3(5)$	4.16, 4.19	4.11
A_7	—	5.6, 5.8

The additional six M_{11} relations of 4.41 are as just specified. Their literal source nodes and transported model types are in `claims.json`; they are not reconstructed from a drawing layout or abstract group identifier during replay.

Proposition 2. *The displayed geometries 3.1, 3.2 and 4.1 are residually primitive for the specified G action: every nonempty type of every residue carries a primitive action of its flag stabilizer.*

Proof. Write $I = \{1, \dots, r\}$, $H_J = \bigcap_{j \in J} P_j$, with $H_\varnothing = G$. Flag transitivity, acquired for each literal geometry, transports every flag of type J to the base flag. For $i \notin J$, its incident objects of type i form the transitive H_J-set with point stabilizer $H_{J \cup \{i\}}$. A transitive coset action is primitive exactly when this stabilizer is a maximal proper subgroup. The statement does not require the action to be faithful: subgroups between a point stabilizer and the acting group correspond to blocks through its base point, and quotienting by the action kernel does not change that interval.

For each required inclusion $L = H_{J \cup \{i\}} < H = H_J$, the certificate retains an index a in the acquired 589-class roster and $x \in G$ with $H^x = K_a$. It also retains an actual maximal-inclusion row (K_a, M) and $y \in K_a$ with $(L^x)^y = M$. Both equalities, both element memberships, and proper containment are verified literally. Since y normalizes K_a, conjugation by xy identifies (H, L) with (K_a, M). The acquired maximality of M in K_a proves the result. The checker enumerates every proper $J \subset I$ and every $i \notin J$, including $J = \varnothing$ and rank-one residues. Hence there are $r2^{r-1}$ inclusions: twelve each for representatives 36 and 35 (diagrams 3.1 and 3.2), and thirty-two for representative 77 (4.1). All fifty-six have positive literal transports. This pays the all-type claim, which is stronger than the acquired weakly primitive condition. □

Proposition 3. *For each of the 86 displayed diagrams, its retained parenthetic number (one when absent) equals the number of acquired inner/type family classes in its fusion fiber under full automorphisms and type permutations. The individual numbers comprise 42 ones and 44 twos, summing to 130.*

Proof. The acquired complete fusion file contains 130 member records with distinct input identifiers and 86 representative identifiers. Its producer explicitly stores the representative as `source` and the original input as `target`. Count records by this source identifier, then apply the accepted source-coordinate bijection. `literal_join.py` requires every identifier, every diagram, and every individual printed number to agree. The fiber counts concern the specified action and its inner equivalence. They are not additional bare incidence classes; the accepted bare classification has 86. □

The finite derivation returned cleanly without an ambient subgroup or geometry census. Its completed rows, exact loaded inputs and terminal receipt are retained. Independent small controls exhaust all subsets of the elements of C_4, D_8, S_3 to form their subgroup sets, test maximality by all intervening subgroups, and test conjugacy by every acting element: 159 comparisons, including negative cases. The replay also rejects damaged source coordinates, fabricated maps, wrong maximal-inclusion indices and reversed fusion directions. An initial control harness emitted unbound-global warnings; its clean successor uses the exact helper prefix without the unused target entry point. A Python symlink-locator mismatch was rejected before the actual call and repaired without changing any mathematical input.

The three-RPRI, 24-conjugacy and 86-number statements are the complete scope of this component. The eight-group comparison table and final source synthesis remain separate, as do already qualified doubling and attribution clauses. No new census total, conjecture, measurement or whole-original admission is assigned here.